

"Heron 2.5: Coil B primary/secondary failure causes cylinder 2 misfire P0302/

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Area: bulletins

1 Condition

Some Heron 2.5 vehicles may exhibit a rough idle, hesitation under load, and a flashing malfunction indicator. The PCM will store DTC P0302 "Cylinder 2 Misfire Detected" together with DTC P0352 "Ignition Coil B Primary/Secondary Circuit" when the cylinder 2 ignition event fails. The simultaneous presence of P0302 and P0352 points specifically at the coil serving cylinder 2 rather than at fuel or compression. Refer to Heron Ignition Service Manual :: Coil Primary and Secondary Circuit Faults for the circuit-level description of these codes.

2 Cause

This condition is caused by an internal winding breakdown in the application-specific ignition coil, part number IC-2044-A, which produces an open or shorted primary/secondary circuit as the coil heats up. The failure of IC-2044-A interrupts the spark event for cylinder 2 and is reported by the PCM as the combined P0302 and P0352 pattern. Coil application and interchange notes are covered in Heron Ignition Service Manual :: Application-Specific Coils .

3 Diagnosis

Before replacing parts, run PROC-MISFIRE-DIAG to confirm the misfire is isolated to cylinder 2 and is not the result of a vacuum leak or fuel concern. Swap the suspect coil to an adjacent cylinder and observe whether the misfire and the P0302/P0352 codes follow the coil, which confirms an internal coil fault. The full diagnostic tree for ignition-side misfires is provided in Heron Ignition Service Manual :: Ignition Circuit Diagnosis and the Heron Ignition Service Manual :: Ignition Coils overview.

4 Repair Procedure

Replace the failed coil with IC-2044-A and inspect spark plug SP-1110 for fouling or erosion, replacing it if it is outside the gap specification. Follow PROC-IGN-COIL and the Heron Ignition Service Manual :: Ignition Coil Replacement Procedure for boot seating and torque values. Verify SP-1110 specifications against Heron Ignition Service Manual :: Spark Plug Selection and Specifications before reinstallation, then clear codes and road test to confirm P0302 and P0352 do not reset. See Heron Ignition Service Manual :: Spark Plugs for related maintenance guidance.